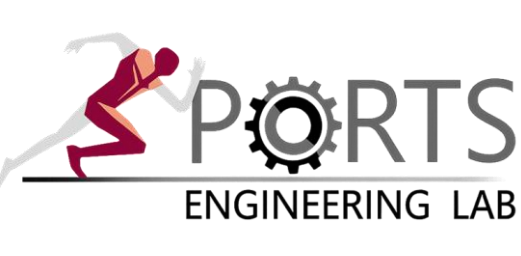


# Referent-Configuration Imitation Learning (RCIL) for Predictive Musculoskeletal Simulation

Faster and more stable gait imitation by controlling referent configurations instead of muscle activations

Chihyeong Lee<sup>1</sup>, Jooeun Ahn<sup>1,2\*</sup>, Beom-Chan Lee<sup>2,3\*</sup>



<sup>1</sup>Department of Physical Education, Seoul National University, South Korea

<sup>2</sup>Institute of Sport Science, Seoul National University, South Korea

<sup>3</sup>Department of Health and Human Performance, University of Houston, USA

\*Corresponding authors

email: [chi0412@snu.ac.kr](mailto:chi0412@snu.ac.kr), [ahnjooeun@snu.ac.kr](mailto:ahnjooeun@snu.ac.kr) and [blee24@central.uh.edu](mailto:blee24@central.uh.edu)

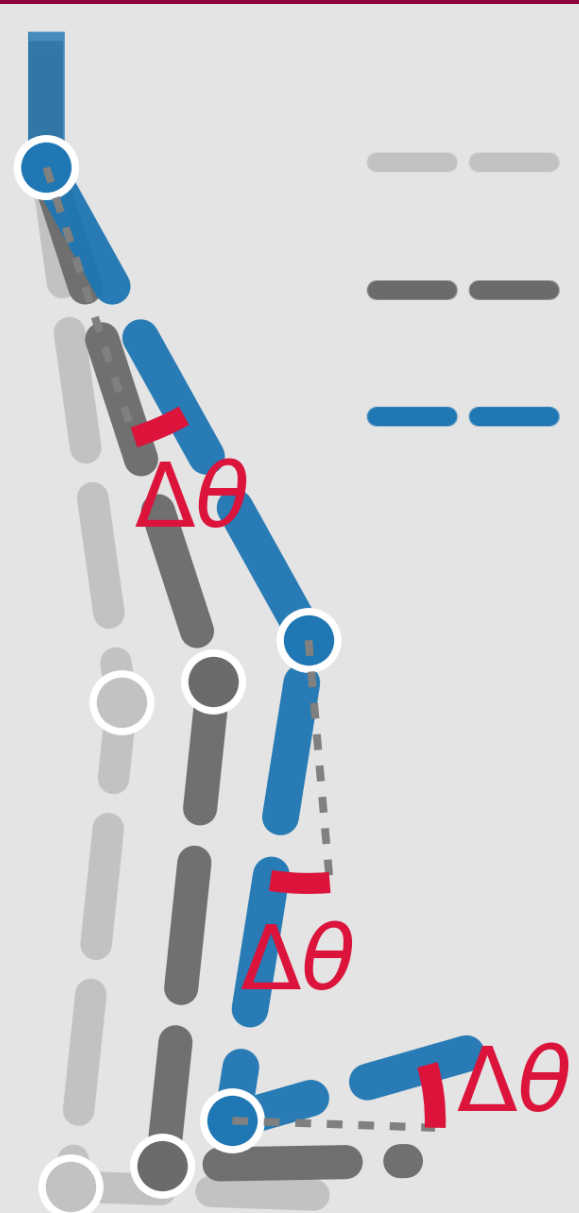
## PROBLEM

- Predictive musculoskeletal simulation (MSK) is powerful, but muscle redundancy makes control hard [1].
- Imitation learning reproduces motion [2], yet controlling in muscle-activation space is slow, unstable, and prone to collapses [3].

## MOTIVATION

- The CNS controls movement by shifting referent configurations ( $\lambda$ -thresholds), rather than muscle activations, a low-dimensional and self-stabilizing control variable [4, 5].
- Our idea: let the policy learn in referent configuration space — a small residual  $\Delta\theta$  added to the reference pose.**
- The 0.1 s anticipatory lead matches the ~100 ms neuromechanical delay that the CNS compensates for [6].

## METHODS & RESULTS



— current  $\theta(t)$   
— reference  $\theta_{ref}(t+0.1s)$   
—  $R = \theta_{ref}(t+0.1s) + \Delta\theta$

$$1) R = \theta_{ref}(t + 0.1s) + \Delta\theta$$

➤  $R$  is the target referent configuration.

$$2) \lambda_R = \text{muscle fiber length}(R)$$

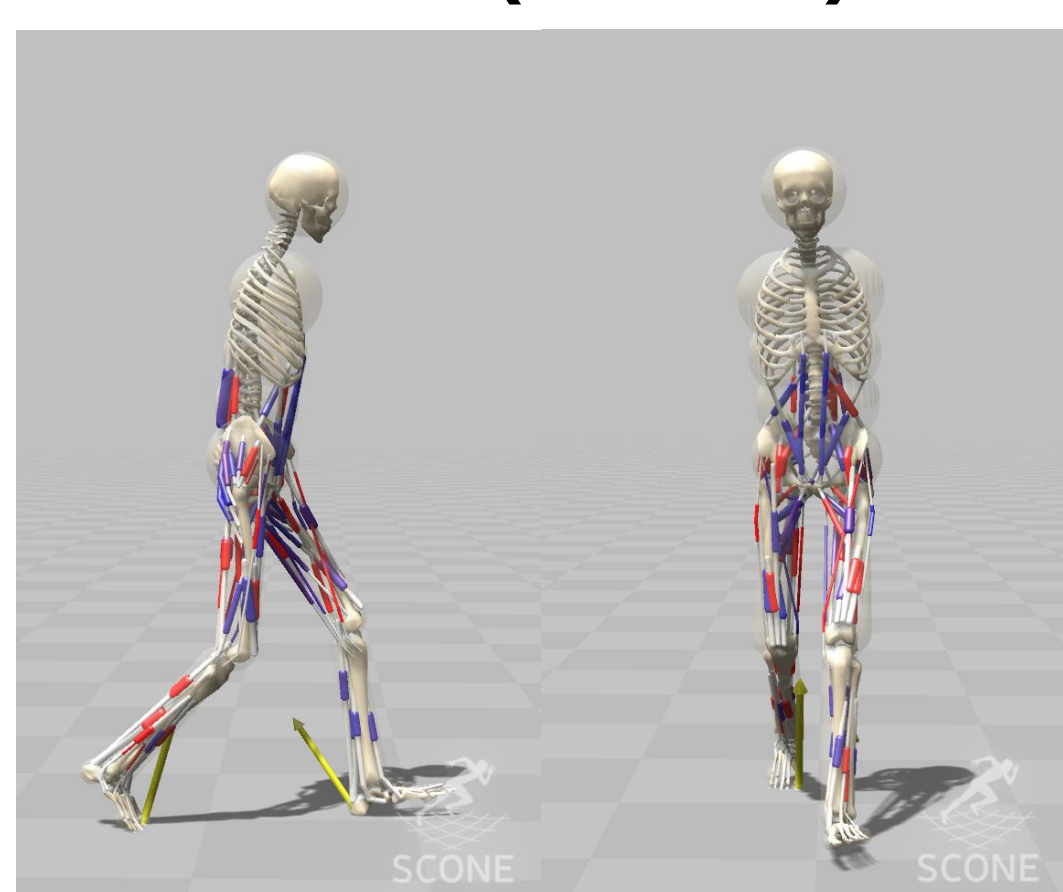
➤  $\lambda_R$  is the muscle fiber length at  $R$ .

$$3) A(t) = [g(L(t) - (\lambda_R - \delta) + \mu V(t))]_0^1$$

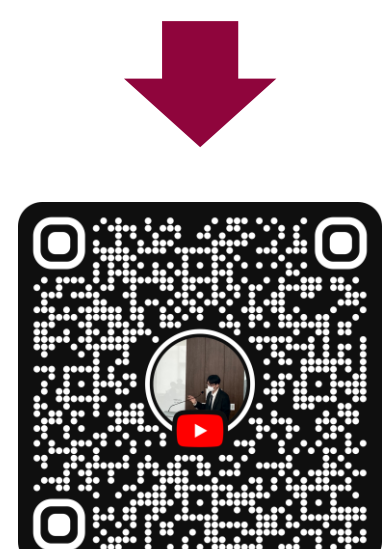
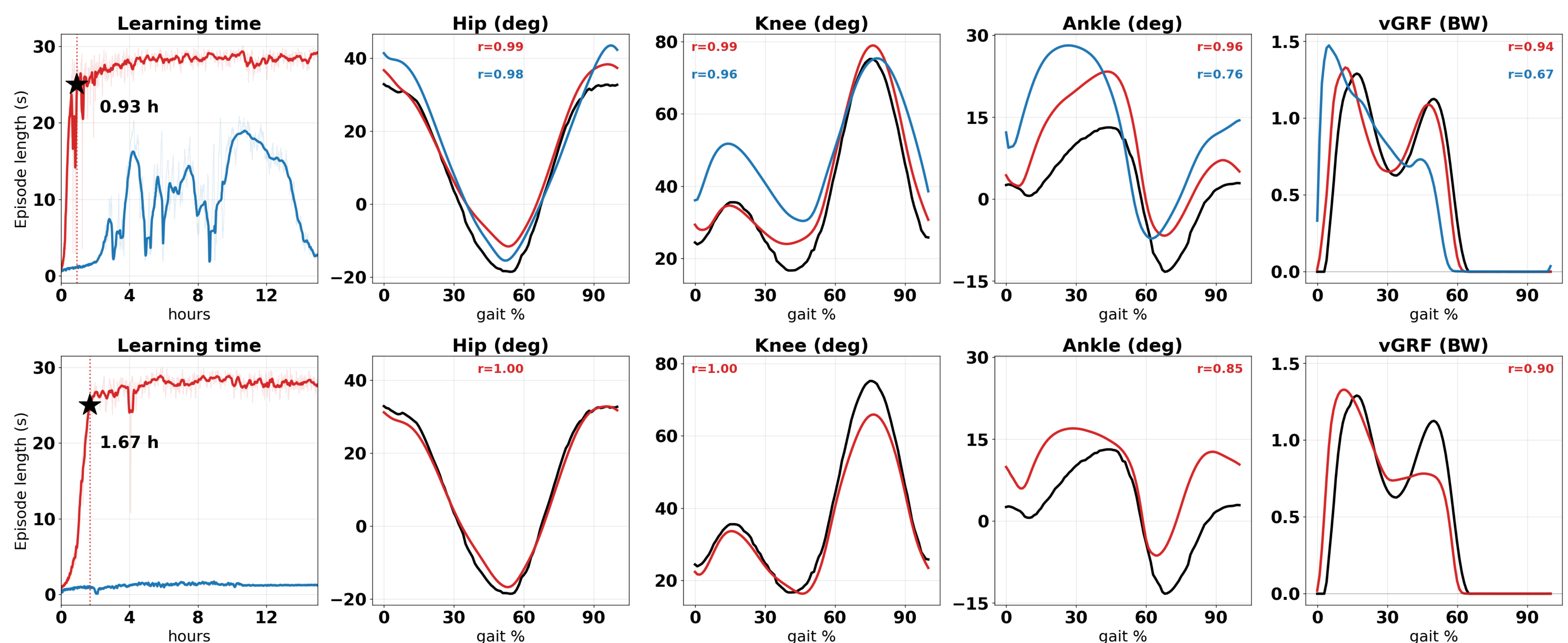
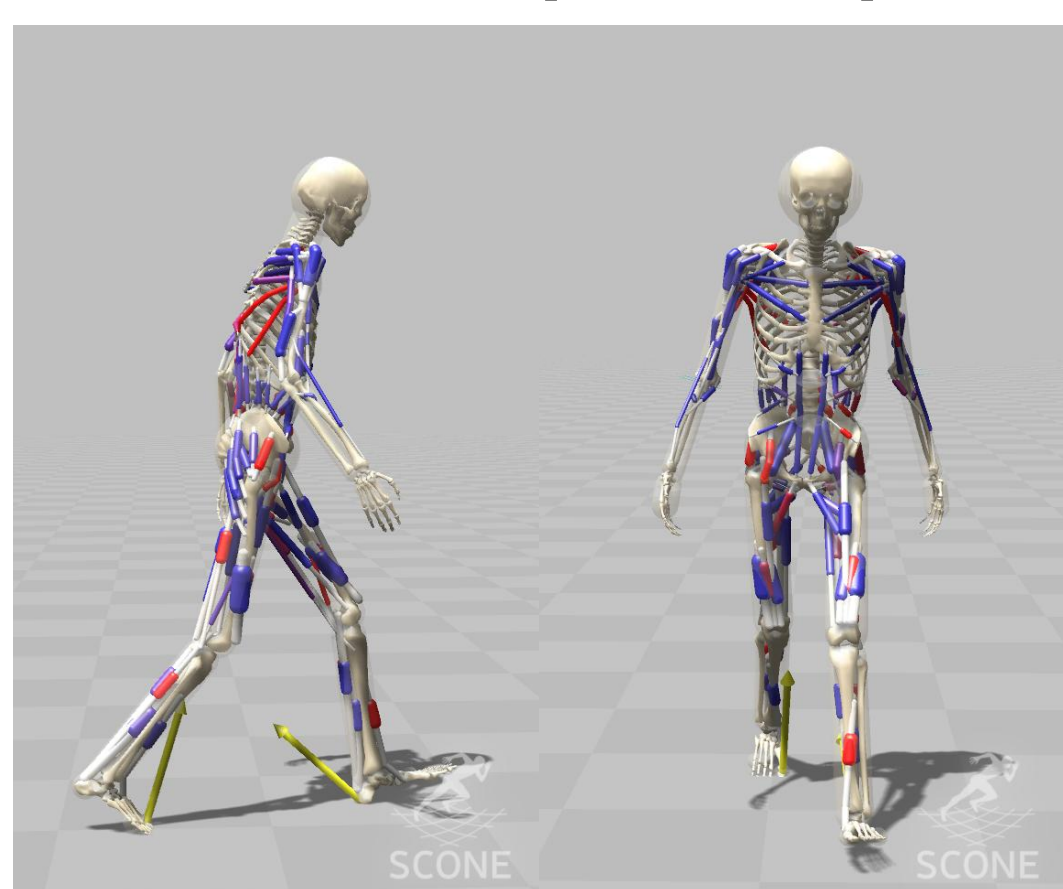
➤  $A(t)$  is the muscle activation [4, 5].

- $L(t)$ : Muscle fiber length
- $[g]_0^{20}$ : Length-based reflex (tonic) gain
- $[\delta]_0^{0.2}$ : Baseline muscle tone
- $V(t)$ : Muscle fiber Velocity
- $[\mu]_{-0.1}^{+0.1}$ : Velocity-based reflex (phasic) gain
- $(\lambda_R - \delta)$ : Length-threshold

H2190 (SCONE)



H32130 (SCONE)



Check this out!

- RCIL reached stable walking within ~1-2 hours in both models (H2190 0.93 h, H32130 1.67 h), whereas pure IL catastrophically forgot (H2190) or never converged (H32130).**
- RCIL tracked reference joint kinematics closely across both models (hip & knee  $r \geq 0.99$ , ankle  $r = 0.96/0.85$ ), while pure IL degraded distally (H2190 ankle  $r = 0.76$ ).**
- RCIL reproduced the vertical GRF ( $r = 0.94/0.90$ ), far above pure IL, which lost push-off (H2190 vGRF  $r = 0.67$ ).**

## DISCUSSION & CONCLUSION

- Commanding referent configurations instead of muscle activations turns muscle redundancy into a low-dimensional, dynamically consistent target, so RCIL converges quickly to stable walking while direct imitation does not.
- RCIL turns motion data into subject-specific musculoskeletal digital twins that are fast to train, accurate, and ready for predictive simulation.

## ACKNOWLEDGEMENT

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## REFERENCES

- [1] Geijtenbeek et al. (2019), SCONE, [2] Peng et al. (2018) [3] Song et al. (2021), [4] Feldman (1986), [5] St-Onge (2004) [6] Afschrift et al. (2021)